

BRUMLEY FINE, MO, USA

WMO: 724459

Lat: 38.096N

Lon: 92.553W

Elev: 265

StdP: 98.18

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 53931

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-14.9	-12.0	-20.6	0.6	-13.6	-17.6	0.8	-9.6	9.3	10.3	8.4	9.7	2.2	340	0.444

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.6	36.0	24.0	33.8	23.9	32.5	23.8	25.9	32.1	25.3	31.7	24.6	30.6	3.5	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
24.1	19.6	29.6	23.2	18.6	28.7	22.7	18.0	28.1	81.6	32.2	78.6	31.5	75.9	30.5	28.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.3	7.2	6.1	DB	-18.6	37.9	3.6	2.2	-21.2	39.5	-23.3	40.8	-25.3	42.0	-27.9	43.6	(4)
(5)			WB	-19.0	27.0	3.4	1.0	-21.5	27.7	-23.5	28.3	-25.4	28.9	-28.0	29.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	13.5	0.3	2.6	7.6	13.7	18.5	23.1	25.5	24.9	20.7	14.1	7.8	2.6
	DBStd	10.17	6.77	6.66	6.25	5.25	4.52	3.38	3.07	3.26	4.15	5.21	5.53	5.79
	HDD10.0	1040	309	222	119	23	2	0	0	0	0	19	107	239
	HDD18.3	2537	561	442	335	156	57	6	1	1	23	150	318	489
	CDD10.0	2314	7	13	46	133	264	392	479	463	321	148	39	9
	CDD18.3	769	0	0	3	16	61	148	222	206	93	20	1	0
	CDH23.3	7652	0	2	33	172	546	1434	2371	2050	858	176	9	0
	CDH26.7	3240	0	0	3	34	159	573	1133	972	323	42	1	0
Wind	WSAvg	2.7	2.9	3.0	3.2	3.4	2.6	2.4	2.1	2.0	2.0	2.4	2.9	2.8
Precipitation	PrecAvg	1159	59	60	86	127	153	123	114	100	120	81	79	60
	PrecMax	1631	185	126	179	402	326	261	288	234	424	263	238	128
	PrecMin	783	14	5	35	20	33	38	25	46	25	9	10	21
	PrecStd	228	39	35	43	73	74	53	68	45	92	56	58	29
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	20.0	22.9	27.1	29.7	32.3	34.6	38.0	37.9	34.2	30.6	24.8	20.3
		MCWB	13.9	14.5	16.6	18.7	22.5	24.5	23.8	23.6	22.5	21.8	16.9	15.3
	2%	DB	17.2	19.1	23.8	27.5	30.2	32.8	36.1	36.1	32.4	27.5	22.0	17.2
		MCWB	11.6	12.6	14.4	17.3	21.8	24.0	24.3	23.8	23.1	19.2	14.4	12.6
	5%	DB	13.7	16.9	21.2	25.8	28.1	32.0	33.9	33.8	30.5	25.1	18.9	14.0
		MCWB	9.1	10.8	13.0	16.3	20.9	23.8	24.4	23.7	22.6	18.0	13.5	9.6
	10%	DB	10.9	13.6	17.9	22.8	27.0	30.2	32.5	32.2	27.9	22.6	17.1	12.0
		MCWB	6.8	8.4	11.3	15.2	19.8	23.1	24.3	23.5	20.9	16.9	12.0	8.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.4	16.1	17.8	20.2	24.2	26.3	26.9	27.1	25.2	22.9	17.8	15.9
		MCDB	19.2	21.1	25.3	27.1	29.6	31.9	33.0	32.6	31.0	28.1	22.0	19.7
	2%	WB	12.5	14.1	15.7	18.9	23.0	25.3	26.0	25.7	24.1	21.2	15.9	13.5
		MCDB	15.7	17.4	21.6	25.3	28.5	31.2	32.5	32.1	29.8	25.1	19.8	16.5
	5%	WB	9.6	11.6	14.0	17.7	21.8	24.4	25.4	25.0	23.2	19.5	14.6	10.8
		MCDB	13.3	15.7	19.5	23.1	27.0	29.9	31.9	31.4	28.7	23.4	18.2	13.3
	10%	WB	6.5	8.5	11.9	16.3	20.5	23.6	24.8	24.1	22.2	17.8	12.6	8.0
		MCDB	10.3	13.5	17.2	22.1	25.2	28.9	31.0	30.0	27.2	21.9	16.5	10.7
Mean Daily Temperature Range	5% DB	MDBR	10.5	11.0	12.2	12.3	11.8	11.2	11.6	11.9	12.1	12.0	11.1	9.9
		MCDBR	14.8	15.6	16.1	15.2	13.2	12.8	13.7	14.6	13.7	13.9	14.2	13.7
		MCWBR	10.2	10.4	9.1	8.1	6.8	5.9	5.0	5.3	6.0	7.4	9.2	9.6
	5% WB	MCDBR	12.7	13.8	13.5	12.4	11.9	11.4	11.7	11.9	11.6	11.5	11.7	11.7
		MCWBR	10.0	10.3	8.6	7.7	6.6	5.9	5.1	5.3	5.7	6.8	8.8	9.4
Clear-Sky Solar Irradiance	taub	0.301	0.313	0.341	0.388	0.421	0.431	0.453	0.457	0.390	0.346	0.309	0.296	
	taud	2.477	2.443	2.418	2.291	2.262	2.265	2.226	2.214	2.373	2.486	2.535	2.517	
	Ebn at Noon	884	918	921	890	863	852	829	816	857	869	870	867	
	Edh at Noon	80	94	107	129	135	135	139	138	111	90	76	72	
All-Sky Solar Radiation	RadAvg	2.22	2.88	3.84	5.00	5.66	6.44	6.31	5.61	4.84	3.48	2.41	1.89	
	RadStd	0.24	0.33	0.33	0.31	0.50	0.52	0.43	0.33	0.34	0.38	0.25	0.20	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (17 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page